

SMTH 322 – Assignment 2

Due Date: 30 July 2026 at 10:00

Submission Venue: Mr Lala's Office

Instructions:

- Answer all questions.
- Show all necessary working.
- Neatness and mathematical justification will be considered.

Question 1

Determine whether $\lim_{z \rightarrow 0} \frac{\bar{z}}{z}$ exists.

Question 2

Consider

$$f(z) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} + i \frac{2xy}{x^2 + y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

Investigate whether $f(z)$ is continuous at the origin.

Question 3

Determine whether the following function is harmonic.

$$u(x, y) = x^2 + y^2 + \ln(x^2 + y^2), (x, y) \neq (0, 0).$$

Question 4

Find the harmonic conjugate of $u(x, y) = e^x \cos y$. Hence express the corresponding analytic function in terms of z .

Good luck!